

ZIXI YI

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Changsha, Hunan, China

Research Interests

Numerical analysis; numerical PDEs; scientific computing; spectral methods; computational geometry and geometric analysis.

Education

Central South University

Sep. 2023–Present

B.S. candidate, Mathematics and Applied Mathematics

GPA: 4.0/4.0 Major rank: 1/54

Selected coursework: Analysis (98/100); Differential Equations (97/100); Scientific Computing (97/100); Differential Geometry (95/100).

Graduate coursework in progress: Functional Analysis; Partial Differential Equations.

Publications and Manuscripts

Preprints and Manuscripts

1. **Z. Yi**, L. Qu, and G. P. T. Choi, “An Integrated Geometric Quantification and Shape Analysis Framework for Axillary Lymph Node Metastasis in Breast Cancer Patients.” *Under review at Biomedical Signal Processing and Control*; arXiv:2609.35437, 2026.
2. **Z. Yi** and G. P. T. Choi, “Conditioning-Aware Surface Parameterization for Finite Spectral Geometry.” *Submitted to arXiv*, 2026.
3. **Z. Yi** and G. P. T. Choi, “Representation-Aware Spherical Parameterization: Exact Sensitivity Analysis and Pole-Aware Planar Preconditioning for FLASH.” *Manuscript in preparation*.
4. **Z. Yi** and Y. Xu, “Numerical Blow-Up in a Rigid-Ignition Model with Non-Fourier Heat Conduction.” *Manuscript in preparation*.

Peer-Reviewed Publications

5. L. Qu, J. Zhu, X. Mei, **Z. Yi**, et al., “Evaluating axillary lymph node metastasis risks in breast cancer patients via Semi-ALNP: a multicenter study,” *eClinicalMedicine* 85 (2025), 103311. DOI: 10.1016/j.eclinm.2025.103311.
6. L. Qu, X. Mei, **Z. Yi**, et al., “An unsupervised learning model based on CT radiomics features accurately predicts axillary lymph node metastasis in breast cancer patients: diagnostic study,” *International Journal of Surgery* 110(9) (2024), 5363–5373. DOI: 10.1097/JS9.0000000000001778.

Research Experience

Conditioning and Sensitivity in Spherical-Harmonic Surface Parameterization

Jun. 2026–Present

Research project with Prof. Gary P. T. Choi, The Chinese University of Hong Kong

- Developing a conditioning-aware framework for finite spectral geometry and tracing perturbations through reconstruction into curvature and Laplace–Beltrami quantities.
- Characterizing exact representation sensitivity and designing pole-aware planar preconditioning for FLASH.

Numerical Blow-Up in a Rigid-Ignition Model with Non-Fourier Heat Conduction

Sep. 2025–Present

Research project with Prof. Yufeng Xu, Central South University

- Analyzing finite-time singularity formation in a nonlinear heat-conduction model; deriving higher-order energy estimates and designing adaptive computation near singular regimes.

Semi-Supervised CT Radiomics for Axillary Lymph-Node Metastasis Prediction

Sep. 2024–Aug. 2025

Research project in medical imaging and machine learning, Central South University

- Built and validated a semi-supervised pipeline on multicenter clinical data, combining radiomic features with machine learning for node-level risk stratification.

Unsupervised CT Radiomics for Axillary Lymph-Node Assessment

Sep. 2023–Aug. 2024

Research project in computational imaging, Central South University

- Extracted 3D radiomic descriptors and applied unsupervised learning to support noninvasive node-level assessment; modeled lymph-node tumor samples using 3D ResNet-50.

Multimodal Intelligent Bird-Recognition and Monitoring System

Apr. 2024–Mar. 2025

Team Leader, Central South University

- Organized and led the project team in developing a protected-area monitoring system; coordinated project planning and implementation across computer vision, edge computing, and audio–visual multimodal data fusion for real-time bird and wildlife recognition.

Furong Laboratory

Aug. 2025–Sep. 2025

Research Intern in Data Analysis

- Cleaned and screened clinical data; compared linear models, evaluated AUC by cross-validation, and created visualization dashboards and interim reports.

Industry and Research Internships

Hengrui Pharmaceuticals, Statistics Department

Jul. 2025–Aug. 2025

Statistical Analysis Intern

- Cleaned and verified clinical data; conducted survival analysis, Cox and logistic regression, and statistical significance testing.

Honors and Awards

Central South University Academic Scholarship, Second Prize

2024–2025

National English Competition for College Students (NECCS), Type C – Second Prize

2024

Central South University Academic Scholarship, Third Prize

2023–2024

Leadership and Service

University-Wide Women’s Running Program

2023

Participant

- Ranked fourth university-wide and first within the college by running volume.

Central South University International Culture Festival

Nov. 2025

Volunteer Organizer and Booth Coordinator, United Kingdom cultural booth

- Coordinated volunteer service at the United Kingdom cultural booth, organized visitor reception and bilingual presentations, and introduced traditional foods and their cultural background in Chinese and English.

Skills and Tests

Programming: Python, MATLAB, C++, R, $\text{\LaTeX}$.
Tests: GRE 324 (Q167, V157); IELTS 7.5.